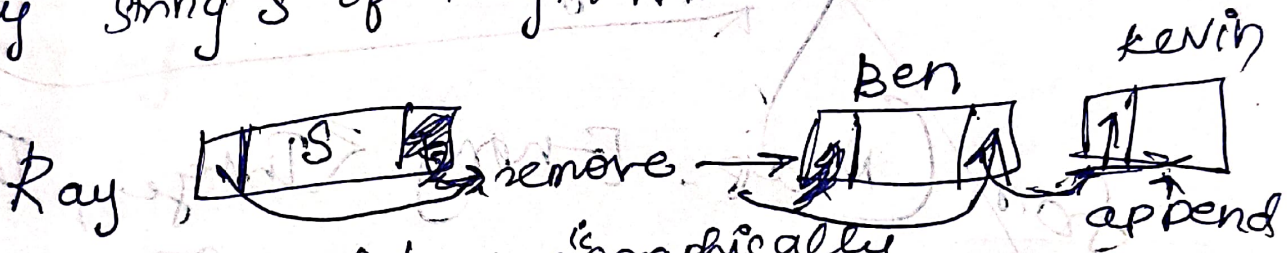


3

MLSS 2023

1. Ray string S of length N .



Goal: Kevin's Lexicographically smallest string should be returned.

Ray Queue F++

Ben stack L++

Kevin o/p

Idea:

For the string in Ray, compute suffix minimum, comparing every character and storing comparison in index.

Then use Ben to push from the original string in the Ray and pop in ascending order, the minimum most-s to Kevin.

1. Sum of Array Array

What we want? sum of gcds of sorted pairs

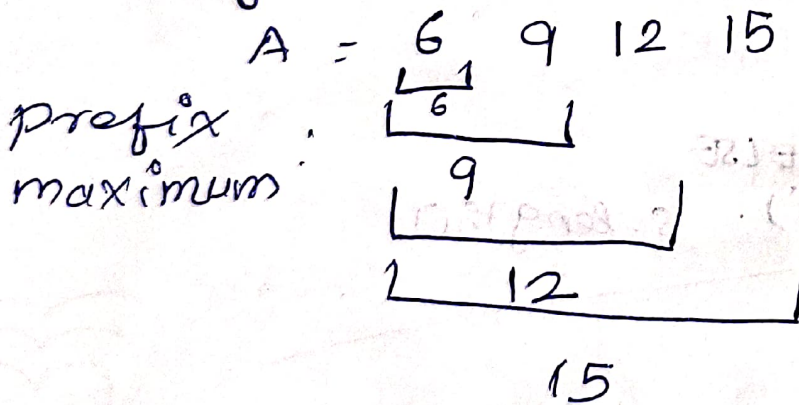
3 concepts:

1. prefix maximum

2. GCD

3. sorting + two pointers Pattern

Toy test case:



$$\text{GCD}(6, 6)$$

$$\text{GCD}(9, 9)$$

$$\text{GCD}(12, 12)$$

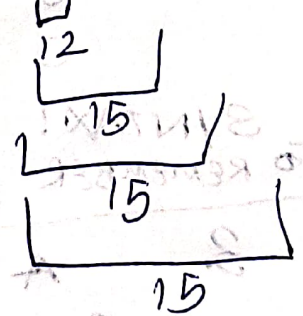
$$\text{GCD}(15, 15)$$

first utility fn which is static

then, helper function for gcd

test case 2

$$A = 12 \ 15 \ 3 \ 10$$



$$\text{GCD}(12, 12) = 12$$

$$\text{GCD}(15, 15) = 15$$

$$\text{GCD}(3, 15) = 3$$

$$\text{GCD}(10, 15) = 5$$

$$\text{GCD}(3, 10) = 1$$

$$\text{GCD}(12, 10) = 2$$

$$\text{GCD}(15, 10) = 5$$

$$\text{sum} = 4$$

2. Key press calculator:

Trick:

How many pairs of consecutive ("00") can you use?

100 2 1 pair

1 0 2 0 no pair

100 00 0 2 2 pairs

5 digits not 7

6 0 0 0 4

↑ ↑ ~~↑~~
↓

+ 1 then skip both zeros and move forward

6 00 0 4
0 1 2 3

3 A B C

3 arrays

XOR

[1 2]

[3]

[4 5]

Triplets: [1 3 4]

[1 3 5] [2 3 4]

[2 3 5]

XOR of 1, 3 and 4 in first set and so on.
convert to binary first? x

inside loop

int xor = a ^ b ^ c

110 → find set bits

even odd

print discard

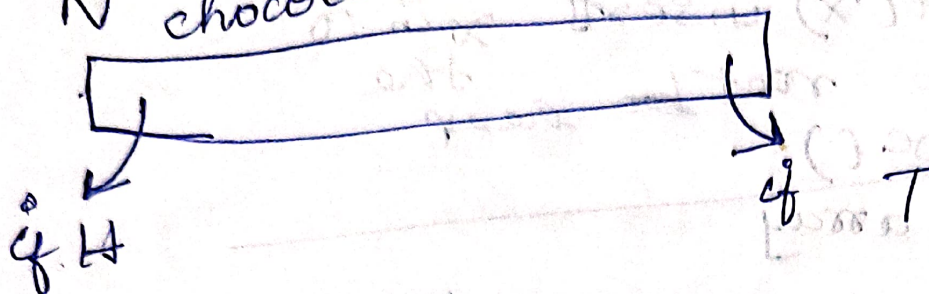
Set Bits = Integer.bitCount(xor)

or n & n-1 of binary

with int can
 $n \leq n-1$

Last chocolate

N chocolates



Find No. of coin-Toss sequences.

$k \rightarrow$ position of last chocolate
after $N-1$ tosses



I/P $N=3$ $k=2$

$N-1$ tosses. After 2 tosses

chocolate at position k remaining

two toss

HT $\rightarrow$ ✓
TH $\leftarrow$ ✓

TT X
HH X

1-based index

clue: we just need to

find

~~NO~~ $N-1 C_{k-1}$